When the Attacker Hides in the Weather: Provable Bounds on Undetected GNSS Spoofing of UAVs

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***Abstract*—A UAV cannot tell from its navigation inputs whether a growing position error comes from a jammer, a drifting sensor or wind, so an attacker can disguise interference as benign. We bound the probability that a mission fails with no detector raising an alarm, for every coupling of the benign disturbances. For any adaptive, fully informed GNSS-offset attacker, a linear programme proves, with finite-sample confidence, the largest undetected deviation of a linear navigation loop. The bound is finite only when a trusted fix re-anchors the estimator, and shrinks from 33.5 m at 5 s to 10.6 m at 1 s. Optimised disguises evade four detectors in 1,909,600 simulated missions, because detectors must tolerate benign noise. In 12,321 PX4 software-in-the-loop missions, the stealth budget transferred but the full disguise matrix did not, and re-anchoring every second kept every undetected deviation within 3.16 m. Engineers and certification authorities can therefore size the trusted re-anchoring interval to the corridor a mission must hold, instead of tuning detector thresholds.**

I. Introduction

An autonomous UAV can be attacked through its command-and-control link, peer mesh, on-board bus and navigation sensors. Each interface has its own defences, with network intrusion detection systems (IDSs) for the links and the bus [1]–[3], physical-invariant and residual monitors for the estimator [4]–[6], and certified robustness for learned components [7], [8]. None of them is designed to show whether, once some attacks get through, the mission can still be provably completed. For a single declared attacker the question has a known answer, restated in Section II-B, in which a Lipschitz–Grönwall tube converts the bounded navigation error caused by a relay delay just below a detector's threshold into a certified floor on the mission-completion rate (MCR). This answer assumes that the attacker is the only source of perturbation.

a) Attacks disguised as benign disturbances.: Real navigation errors have many benign causes. A GNSS receiver drifts with the ionosphere and multipath, an accelerometer bias steps or random-walks, and wind pushes the airframe. Every monitor must therefore tolerate these disturbances, and that tolerance is the space in which an attacker can hide. Control-invariant monitors must accommodate deviation due to friction or wind [6], and an actuator attack lying in the range of the fault-input matrix is indistinguishable from a fault [9]. A GPS-only attacker can cause arbitrarily large deviation against any detector when the closed loop is incrementally exponentially stable [10], and a recent spoofer exploited a PX4 vehicle's wind compensation to deviate it undetected [11]. From the navigation input alone, shaped interference, a sensor fault and an environmental disturbance can thus be indistinguishable, and an attacker will choose the disguise that the defender models least well. A certificate gated on detector silence is then unsound against an attacker who dresses as the weather. A useful guarantee must instead be source-agnostic, bounding the mission outcome whatever mixture of attack, fault, outage and wind produced the perturbation, under any coupling.

b) What becomes possible.: Spoofing detectors give an alarm, but no statement about the mission while the alarm stays silent. Stealthy-attack reachability [12], [13] bounds the error an attacker can cause under one residual detector, not under a stack of heterogeneous detectors with benign disturbances of unknown dependence, and not as a probability of mission failure. Control-invariant detection [4], [5] measures what an attacker achieves undetected, without a proof for an adaptive attacker who chooses the disguise. Runtime assurance [14] switches to a safe controller once a monitor fires, and gives no guarantee while the monitor is fooled. We bound the probability of undetected mission failure for a stack of detectors under every coupling of the benign disturbances, prove the undetected deviation for an explicit adaptive attacker class, and show that the re-anchoring interval, rather than any threshold, is the design parameter with a proved consequence.

c) Approach and contributions.: We therefore replace the detection gate with a unified perturbation budget and test the resulting bound against attackers optimised to evade it.

- **A threat model of adversarial equivalence (Section III).** We define ε-observational equivalence, bound any detector's advantage by the total-variation (TV) distance (Lemma 1), and show that an equivalent attack can violate a detector-gated certificate (Proposition 1). Unlike earlier work on fault mimicry [9], [15], [16], we let the attacker optimise the disguise.
- **A unified budget with mixed semantics (Section IV).** $\delta_{\text{tot}} = \delta_{\text{net}}(\theta) \oplus \delta_{\text{adv}}(\theta) \oplus \delta_{\text{gnss}} \oplus \delta_{\text{fault}} \oplus \delta_{\text{env}}$ combines deterministic, stochastic and probability-box (p-box) terms. We prove a dependence-sharp bound on the probability of undetected mission failure (Theorem 1) and show when the union bound [17], [18] is sharp and when it is loose (Proposition 2).
- **A proved stealth budget (Section IV-E).** For every causal, adaptive GNSS-offset attacker, a linear programme bounds the undetected deviation; the bound is finite only with trusted

re-anchoring and falls from 33.5 to 10.6 m as the interval shrinks from 5 to 1 s.

- **An evasion study and a soundness test (Sections VI and VII).** We cross-evaluate attackers optimised over five disguises against four detectors, and test the bound on held-out attackers with negative controls.
- **A negative result and its repair (Sections VII-E and VII-F).** Stronger optimisers and a test scaled to 1,620 points found the pilot instantiation unsound, with 5 Holm-significant violations traced to two unsecured conditions unrelated to disguise. A run-time saturation alarm and a finite-sample p-box repair both.
- **A pre-registered PX4 transfer study (Section VII-H).** In 12,321 SITL missions with EKF2, the stealth budget transferred, the pre-registered transfer of the full disguise matrix did not, and re-anchoring every second kept every undetected deviation within 3.16 m.

d) Scope.: The main evaluation uses a reduced-order lateral multirotor model with a Kalman filter (KF), and the proved budget holds for that linear loop. Section VII-H repeats the key experiments in PX4 software-in-the-loop (SITL) simulation with its EKF2 estimator under a pre-registered protocol. Throughout, "certified" means proved under the conditions stated with the result.

II. Preliminaries and Problem Statement

A. Navigation loop and mission completion

Let $x(t) \in \mathbb{R}^n$ be the navigation state of the aircraft and $\dot{x} = F(x,u)$ its closed-loop dynamics under feedback u computed from the estimated state $\hat{x}$. A mission ω of horizon T follows a nominal path x^{nom} inside a corridor of half-width (margin) m. Writing $e(t)$ for the cross-track deviation of the true position from the nominal path, the mission completes if and only if

$$\sup_{t\in[0,T]} |e(t)| \le m. \tag{1}$$

The mission-completion rate is $\text{MCR} = \Pr_{\omega\sim\mathscr{M}}[(1)]$ for a stated mission distribution $\mathscr{M}$; it is a population quantity, not a statement about any single flight. Section A collects the notation.

B. Background: A certificate for delayed navigation input

This subsection states the certificate for a single attack class on which the network term of our budget rests, together with its constants. The constants are computed in the released artifact [19]: the rate γ from the real GPS jamming and spoofing flights of the UAV Attack Dataset [20], and the slack s and the local Lipschitz estimate L in the artifact's relay and dynamics analyses.

Tube. Suppose F is L-Lipschitz on the operating region (condition C1), and the filter re-anchors on an uncorrupted update at least once every certification horizon T_c (condition C2). If a perturbation leaves an error of at most δ in the state at the start of a window, the error then obeys $\|\dot{e}\| \le L\|e\|$ with $\|e(0)\| \le \delta$, and Grönwall's inequality [21] bounds the deviation over the window by the tube radius $\rho = \delta e^{LT_c}$.

Hidden delay. A multi-hop forwarding detector flags any hop whose forwarding delay exceeds a per-hop threshold θ, and a message that misses its contact window, with slack s, is flagged regardless of the threshold. An attacker controlling H relays on a route can therefore hide at most $\min(\theta,s)$ of delay per hop, and at most $\Delta(\theta) \le H\min(\theta,s)$ of staleness in total. In the released detector replay, no undetected per-hop delay exceeded 4.84 s over 15,392 relay hops, below the 5 s contact-window slack, and we use this measured cap for s.

Delay-to-position rate. While the estimator waits for a delayed message, its position error grows at a rate γ, evaluated at the supremum over 674 post-onset samples of the flights above, $\gamma = 1.625$ m/s. The network contribution to the deviation is therefore at most

$$e^{LT_c}\,\delta_{\text{net}}(\theta), \qquad \delta_{\text{net}}(\theta) = \gamma H \min(\theta,s), \tag{2}$$

with $L = 1.181$ and $T_c = 1$ s. At the operating point used throughout ($\theta = 0.25$ s, $H = 2$) this contributes 2.65 m.

C. Perturbation sources and their semantics

Five sources perturb the navigation loop. They differ in magnitude and in the kind of statement that can be made about them.

Term	Origin	Semantics
$\delta_{\text{net}}(\theta)$	relay delay below θ	deterministic worst case
$\delta_{\text{adv}}(\theta)$	disguised GNSS offset below the detector stack at θ	deterministic worst case (stealth budget)
δ_{gnss}	noise, Gauss–Markov drift, outages, jamming noise	stochastic, known marginal
δ_{fault}	inertial bias steps and drift, actuator loss of effectiveness	stochastic, known marginal
δ_{env}	turbulence within a declared intensity range	bounded-unknown (p-box)

A deterministic term is a number that holds for every admissible behaviour. A stochastic term has a cumulative distribution function (CDF) that we estimate on a calibration split. A bounded-unknown term has a distribution known only to lie between a lower and an upper CDF, a probability box [22]. For wind, the box is spanned by the Dryden turbulence model of MIL-F-8785C [23] between light and moderate intensity. The sources do not, however, come with a joint distribution. Nature couples gusts and outages in ways that have not been measured, and an attacker who induces jamming can choose when GNSS degradation coincides with gusts.

D. Problem statement

Given (i) the marginal laws of the stochastic sources, (ii) an envelope for the bounded-unknown source, (iii) a detector stack operating at false-positive rate (FPR) budget α with thresholds θ, and (iv) an attacker free to choose both the shape of its interference and its coupling to the benign sources, we seek:

P1 an upper bound on the probability that a mission fails and its attack goes undetected, valid for every coupling of the benign sources and every attacker, i.e., a source-agnostic bound;

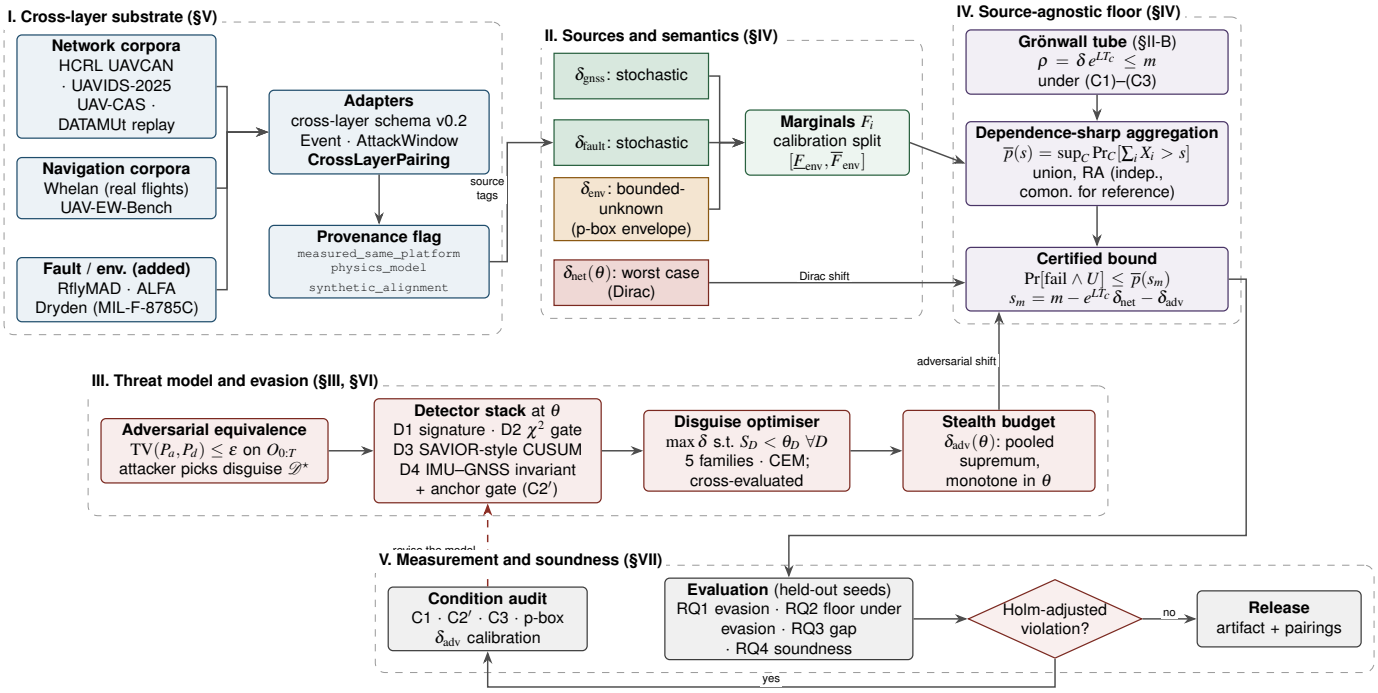


Fig. 1. Methodological framework. Phase I is the experimental substrate: network and navigation corpora under a cross-layer schema whose unit of record is the cross-layer pairing, extended here with fault and environment sources; every pairing carries a mandatory provenance flag. Phase II assigns every perturbation source its probabilistic semantics and estimates its marginal on a calibration split. Phase III formalises adversarial equivalence and searches, for each detector stack and operating point θ, for the disguise that maximises undetected deviation; the pooled supremum is the stealth budget $\delta_{\text{adv}}(\theta)$. Phase IV aggregates the stochastic and bounded-unknown terms with dependence-sharp bounds and shifts the result by the deterministic terms, yielding a certified floor that does not depend on which source produced the perturbation. Phase V tests that floor on held-out attackers and rollouts; a Holm-significant violation triggers an audit of the stated conditions rather than a re-tuning of constants (dashed feedback).

P2 a characterisation of how much tightness each aggregation rule sacrifices and where the slack comes from;

P3 a measurement substrate that keeps the evidence behind every coupling explicit, and an evasion study that measures which disguise defeats which detector.

These define seven research questions. RQ1: can an attacker mimic a fault, a GNSS drift or a gust well enough to evade a detector while still producing a large deviation, and which disguise does each detector miss? RQ2: does the source-agnostic floor bound the outcome of missions whose attack went undetected? RQ3: how loose are the union bound and the dependence-sharp bound in practice, and is the looseness due to dependence or to time? RQ4: is the floor sound at every held-out operating point under Holm-corrected tests, and does the test detect unsoundness when a condition is deliberately broken? RQ5: does the answer to RQ4 survive stronger attackers and a larger test, and if not, which condition fails? RQ6: how much does the floor change when the simulated GNSS marginal is replaced by measured jamming data? RQ7: does the result carry over to a full autopilot estimator, and does the re-anchoring rule hold there?

Figure 1 summarises the methodology through which the paper answers these questions.

III. Threat Model: Adversarial Equivalence

A. Capabilities, knowledge and goal

The attacker can (a) delay relayed messages on up to H hops of the mesh, as in Section II-B, and (b) add an offset $d(t)$ to the GNSS position the navigation filter receives. Capability (b) is realised in practice by signal-level spoofing [24]–[26]; in this paper it is exercised only through simulator interfaces. The attacker cannot corrupt the inertial measurement unit (IMU), and it cannot corrupt a trusted re-anchoring fix, such as a visual landmark or an authenticated position broadcast, that arrives at least once every T_a seconds (condition C2′, which generalises the re-anchoring condition C2 of Section II-B from the network channel to the GNSS channel). We return to IMU injection [27], [28] in Section VIII.

Following Kerckhoffs's principle, the attacker knows the detector designs and their thresholds. It also observes the local wind, which a co-located attacker can measure. It does not know the realisation of the benign noise. Its goal is to maximise the vehicle's cross-track deviation while raising no alarm at any time during the mission. Being detected late counts as being detected.

B. Observational equivalence

Let $O_{0:T}$ denote everything the defender observes: GNSS fixes, IMU samples, commanded accelerations and anchor fixes. Write P_a for the law of $O_{0:T}$ under attack a (with the benign

sources present), and P_d for its law under a benign disturbance d drawn from a class $\mathscr{D}$, such as inertial-bias drift, GNSS multipath drift, or turbulence of a given intensity.

Definition 1 (ε-observational equivalence)**.** *Attack a is ε-equivalent to the class $\mathscr{D}$ if $\inf_{d\in\mathscr{D}} \mathrm{TV}(P_a,P_d) \le \varepsilon$, where $\mathrm{TV}(P,Q) = \sup_A |P(A)-Q(A)|$ over measurable events A.*

Lemma 1 (Detection advantage; Le Cam)**.** *For every detector $\psi : O_{0:T} \to \{0,1\}$ and every $d \in \mathscr{D}$, $\Pr_a[\psi=1] \le \Pr_d[\psi=1] + \mathrm{TV}(P_a,P_d) \le \Pr_d[\psi=1] + \sqrt{\mathrm{KL}(P_a\|P_d)/2}$.*

Proof. The event $\{\psi=1\}$ is measurable, so $|P_a(\psi=1) - P_d(\psi=1)| \le \mathrm{TV}(P_a,P_d)$ by definition. The second inequality is Pinsker's [29]. □

Lemma 1 unifies three earlier notions. The "hidden" attacks of Hashemi et al. raise alarms at the false-alarm rate [16]; the ε-stealthy attacks of Bai et al. bound the Kullback–Leibler (KL) divergence [15]; and the fault-mimicking attacks of Roy and Dey are the noiseless case $\varepsilon = 0$ in which $\mathscr{D}$ is a fault class [9]. A detector with false-alarm rate α on $\mathscr{D}$ thus catches an ε-equivalent attack with probability at most $\alpha+\varepsilon$, whatever its design.

Definition 2 (Disguise-optimal attacker)**.** *For a detector stack $\Psi_\theta = \{\psi_{D,\theta_D}\}_D$ and disguise families $\{\mathscr{D}_k\}$, each parameterised by $\phi \in \Phi_k$, the attacker solves*

$$\delta_k^\star(\theta) = \sup_{\phi\in\Phi_k} \mathbb{E}\big[\Delta_\phi \cdot \mathbb{1}\{\psi_{D,\theta_D}(O^\phi) = 0 \ \forall D\}\big], \tag{3}$$

where $\Delta_\phi = \sup_t |e(t)|$ is the induced deviation. It then selects $k^\star(\theta) = \arg\max_k \delta_k^\star(\theta)$, so that $k^\star$ is the disguise class the stack tolerates most.

Proposition 1 (Detector-gated certificates and equivalence)**.** *Let ψ have false-alarm probability at most α under every $d \in \mathscr{D}$, and let $f_{\mathscr{D}}$ be a floor valid under $\mathscr{D}$. Let a be ε-equivalent to $\mathscr{D}$ with $\alpha+\varepsilon < 1$, and let $q_a = \Pr_a[\mathit{fail} \mid \psi = 0]$. Then*

$$\mathrm{MCR}_a \ \le\ 1 - q_a(1-\alpha-\varepsilon).$$

Hence $f_{\mathscr{D}}$ is violated by a whenever $q_a(1-\alpha-\varepsilon) > 1 - f_{\mathscr{D}}$.

Proof. By Lemma 1, $\Pr_a[\psi=0] \ge \Pr_d[\psi=0] - \varepsilon \ge 1-\alpha-\varepsilon$. Then $1-\mathrm{MCR}_a \ge \Pr_a[\mathrm{fail} \wedge \psi = 0] = q_a \Pr_a[\psi=0]$. □

A floor that is to remain valid against every such attack must therefore bound the deviation the attack itself causes on undetected missions, because detector silence is not evidence of a benign disturbance. The proposition formalises the statement that "the decisive attack is the one that looks benign". Without condition C2′, Khazraei et al. show that a GPS-only attacker can make the undetected deviation grow without bound in an incrementally exponentially stable closed loop [10], so q_a approaches one for any fixed corridor. With C2′, the deviation an undetected attacker can accumulate is limited by the anchor interval. The stealth budget $\delta_{\mathrm{adv}}(\theta)$ of Section IV is our estimate of that limit.

IV. The Unified Budget and the Source-Agnostic Floor

A. From sources to an additive budget

For each benign source $i \in \{\mathrm{gnss}, \mathrm{fault}, \mathrm{env}\}$, let $e_i(t)$ be the deviation the closed loop would exhibit under source i alone, and define its *sup-deviation* $X_i = \sup_t |e_i(t)|$. Let e_{adv} be the deviation under the attack alone.

Lemma 2 (Sub-additivity for additive inputs)**.** *Fix the time-varying parameters of a mission (GNSS outage schedule, actuator effectiveness, anchor schedule). If the closed loop is then linear in its additive inputs (wind force, sensor noise and bias, attack offset), the deviation decomposes as $e = e_{\mathrm{ben}} + e_{\mathrm{adv}}$, and $\sup_t |e(t)| \le \sup_t |e_{\mathrm{ben}}(t)| + \sup_t |e_{\mathrm{adv}}(t)|$.*

Proof. Superposition for a linear time-varying system, then the triangle inequality. □

Two remarks limit the scope of the lemma. First, actuator saturation is not linear, so the lemma holds only inside the region where (C1) holds, and (C1) excludes saturation from that region for this reason. Second, outages and loss of effectiveness are *parametric* sources: they change the system rather than add an input. The further decomposition of e_{ben} into per-source suprema, $\sup_t |e_{\mathrm{ben}}| \le \sum_i X_i$, is therefore an assumption, which we state as condition (C5) and test mission by mission in Section VII.

B. Dependence-sharp aggregation

Because the X_i share no known joint law, we bound the tail of their sum over all couplings consistent with the marginals. For non-negative $X_1,\dots,X_d$ with survival functions $S_i(t) = \Pr[X_i > t]$, define the worst-case and standard bounds

$$\overline{p}(s) = \sup_C \Pr_C\Big[\textstyle\sum_i X_i > s\Big], \tag{4}$$

$$p_U(s) = \min\Big\{1,\ \inf_{s_1+\cdots+s_d=s} \textstyle\sum_i S_i(s_i)\Big\}, \tag{5}$$

where the supremum runs over all copulas C.

Proposition 2 (Tightness of the union bound)**.** *(a) $\overline{p}(s) \le p_U(s)$ for all s. (b) For $d=2$, $\overline{p} = p_U$ at every continuity point: the union bound is sharp [17], [18], [30]. (c) For $d \ge 3$ the inequality can be strict. (d) Against a known benign coupling C_0, the ratio $p_U(s)/\Pr_{C_0}[\sum X_i > s]$ never exceeds d under comonotonicity (take $s_i = F_i^{-1}(u)$) and equals d for identically distributed comonotone sources whenever the equal split is optimal. Under independence it is unbounded: for two independent $\mathrm{Exp}(1)$ sources, $p_U(s) \approx 2e^{-s/2}$ while the truth is $(1+s)e^{-s}$. (e) A deterministic term $X_j \equiv c$ has a single coupling with any other variable, so it shifts s by c and contributes no dependence gap.*

Proof. (a) For any split, $\{\sum X_i > s\} \subseteq \bigcup_i \{X_i > s_i\}$. (b) is Makarov's theorem [17], proved independently by Rüschendorf [18]; Frank et al. give the attaining coupling [30]. (c) Embrechts et al. give examples of strict inequality [31]. (d) is direct computation; for independent exponentials, p_U is

attained at the equal split and the ratio $2e^{s/2}/(1+s)$ diverges (Section B). (e) The Fréchet class of a Dirac marginal is a singleton. □

For $d \geq 3$, the Rearrangement Algorithm (RA) approximates $\overline{p}$ with excellent empirical accuracy but without a general convergence proof [31], [32].

a) Control of the coupling.: The adversarial offset enters as a deterministic term, so by (e) it has no coupling to choose. The dependence-sharp bound is needed for two other reasons. First, nature's coupling of gusts, outages and faults has never been measured. Second, an attacker who controls a stochastic source, for example by jamming to induce the outages and noise of δ_{gnss}, chooses when that source fires relative to the others, and hence its coupling, provided the induced episodes leave the marginal of X_{gnss} within its calibration (C3). A common intuition holds that a joint treatment recovers the union bound's slack precisely when the sources are comonotone. Proposition 2 shows otherwise. When the coupling is unknown or adversarial, the union bound is already the right quantity for two sources and nearly right for three, and slack exists only relative to a coupling that is known to be benign.

C. The floor

Theorem 1 (Source-agnostic, dependence-sharp bound)**.** *Assume (C1) a Lipschitz, unsaturated closed loop; (C2′) a trusted re-anchoring fix at least every T_a seconds, including at the end of the mission; (C3) transfer of the calibrated interface quantities (γ, the marginals of X_i, and the stealth budget δ_{adv}) to the deployment; (C4) an environmental marginal in the declared envelope, with CDFs $\underline{F}_{\text{env}} \leq F_{\text{env}} \leq \overline{F}_{\text{env}}$; (C5) subadditivity across benign sources; and (C6) additivity of the network contribution, bounded by $e^{LT_c}\delta_{\text{net}}(\theta)$, to the others. Let U be the event that the stack Ψ_θ raises no alarm. Then, for every coupling of the benign sources and every attack,*

$$\Pr[\mathit{fail} \wedge U] \;\leq\; \overline{p}\big(s_m\big), \qquad s_m = m - e^{LT_c}\delta_{\text{net}}(\theta) - \delta_{\text{adv}}(\theta), \tag{6}$$

with $\overline{p}$ evaluated at the stochastically largest environmental marginal, whose CDF is $\underline{F}_{\text{env}}$. Consequently $\Pr[\mathit{complete} \mid U] \geq 1 - \overline{p}(s_m)/\Pr[U]$*, and in the population of all missions* $\Pr[\mathit{fail} \wedge U] \leq 1 - f(m)$ *with* $f(m) = 1 - \overline{p}(s_m)$.

Proof. On U, (C3) gives $\sup_t |e_{\text{adv}}| \leq \delta_{\text{adv}}$, and (2) bounds the network contribution by $e^{LT_c}\delta_{\text{net}}(\theta)$, which adds to the others by (C6) (Lemma 2 does not cover the network term). By Lemma 2 and (C5), failure on U implies $\sum_i X_i > s_m$. Hence $\Pr[\text{fail} \wedge U] \leq \Pr[\sum_i X_i > s_m] \leq \overline{p}(s_m)$, which needs no conditioning on U. The tail of a sum is non-decreasing under first-order stochastic dominance of any summand, for every coupling and hence for the supremum, so the stochastically largest member of the p-box is the worst (Section B). The conditional form follows by dividing by $\Pr[U]$. □

Remark 1. The theorem is conditional in the same sense as the delay certificate of Section II-B: its first and last links are proved, and its interface quantities are measured. We therefore test (C3)–(C5) empirically in Section VII instead of asserting them. For the adversarial input of (C3), the proved budget of Section IV-E can replace the measured one; it holds for every attacker in its class on the linear loop. Bounding the joint event $\{\text{fail} \wedge U\}$, rather than the completion rate conditional on silence, allows the bound to use unconditional benign marginals, since conditioning on silence would select benign realisations and change their law.

D. Computing the floor

The marginals are estimated on a calibration split (seed block 1xxx) from 4,000 missions per source. p_U is minimised over a grid of splits. $\overline{p}$ is approximated by the RA with $N =$ 1,000 quantile levels, using both the upper and the lower discretisation, keeping the larger value, and bisecting on the confidence level [31], [33]; Section C gives the procedure. p_U is a proven upper bound on $\overline{p}$; the RA value, with the conservative discretisation, is a numerical approximation of $\overline{p}$ that is not proven in general. We report $\min(\text{RA}, p_U)$ as our estimate of $\overline{p}$, which also removes the RA's $1/N$ resolution floor in the far tail, and we give p_U alongside wherever a bound must be strictly certified. For reference only, we also report the tail under independence (Monte Carlo) and under comonotonicity (quantile sum); neither is a certificate.

E. A proved stealth budget for an explicit attacker class

Theorem 1 takes δ_{adv} as an input. An empirical supremum over optimised attackers is only a lower bound on it (Section VII-E), so we also prove an upper bound for a stated class of attackers.

Definition 3 (Attacker class $\mathscr{A}_{\text{GNSS}}$)**.** *An attacker in $\mathscr{A}_{\text{GNSS}}$ chooses one GNSS position offset d_j per GNSS epoch, causally and adaptively, with full knowledge of the model, the noise realisation and every threshold. It cannot alter the IMU, the actuators or the trusted re-anchoring fix.*

Proposition 3 (Uniform proved budget)**.** *On the linear loop of Section VI, on a mission with constant loss of effectiveness and no GNSS outage, let Ψ contain D1, the anchor gate and the saturation alarm D_{sat}. Let each of the m benign margins be the order statistic of N calibration missions that upper-bounds its $1-\varepsilon/m$ quantile with confidence $1-\delta/m$. Then, with confidence at least $1-\delta$ over the calibration sample and probability at least $1-\varepsilon$ over the benign inputs, every attack in $\mathscr{A}_{\text{GNSS}}$ that leaves Ψ silent has $\sup_t |e_{\text{adv}}(t)| \leq \delta_{\text{cert}}(T_a)$, the value of the linear programme of Section VII-F.*

Proof. By superposition (Lemma 2, exact on the event U because D_{sat} is silent), the attack-only deviation and the attack-only parts of the detector statistics are linear in d. Silence of each detector in Ψ, together with the benign margins, implies a convex (here polyhedral) constraint on d. The bound holds for every d in the feasible set, which contains every sequence an adaptive policy can produce on that realisation. The order statistic $X_{(k)}$ falls below the q-quantile only if at least k of the N samples do, with probability $P(\text{Bin}(N,q) \geq k) \leq \delta/m$ for

TABLE I

PROVED WORST-CASE UNDETECTED DEVIATION δ_{cert} (M) FOR $\mathscr{A}_{\text{GNSS}}$ AGAINST THE RE-ANCHORING INTERVAL T_a: D1, ANCHOR GATE AND D_{sat}, 1% FPR, $\varepsilon = 0.01$, LINEAR LOOP, WORST OVER LOSS OF EFFECTIVENESS; MARGINS AT LARGE-SAMPLE QUANTILES AND WITH THE FINITE-SAMPLE CORRECTION (CONFIDENCE $1-\delta$, $\delta = 0.05$).

T_a (s)	large-sample	corrected
0.5	7.5	7.8
1	10.2	10.6
2	14.8	15.5
3	19.7	20.6
5	31.9	33.5
10	77.7	81.8

any continuous distribution; union bounds over the m margins give the confidence and the probability. □

Finite only with trusted re-anchoring. At $T_a = 5$ s and 1% FPR the programme is unbounded with the anchor gate alone (an attacker that carries the estimate along with the vehicle passes every fix), 319.9 m with D1 added, and 31.9 m with D_{sat}. (These values use large-sample margins.) Without the trusted fix it is unbounded with D1 and 1,374 m with D_{sat}, limited only by the mission length. A useful worst case for $\mathscr{A}_{\text{GNSS}}$ thus needs both the fix and the saturation alarm.

The benign envelope is the attacker's budget. D3's alarm condition is also convex: its CUSUM stays below θ_3 if and only if every window sum of squared residuals stays below $\theta_3 + (1+\kappa)|W|$, which with the triangle inequality gives second-order-cone constraints. With the margins split over these constraints as well, adding them moves the bound only from 35.9 to 35.2 m (at $t = 58.4$ s), because $\theta_3 \approx 2{,}479$ must admit benign jamming noise and outages. D2 and D4 alarm on runs of exceedances, which need mixed-integer constraints; omitting them keeps the bound valid.

A design rule with a proof. Table I gives the proved budget against the re-anchoring interval. It falls almost linearly, so re-anchoring every second certifies a 20 m corridor against every attacker in $\mathscr{A}_{\text{GNSS}}$, and every 0.5 s a 10 m corridor, before the network tube and the benign floor of Theorem 1 are subtracted. The scope is the linear loop. The margins are distribution-free (the 6th largest of 4,000 calibration values, $\delta = 0.05$) and raise the bound by at most 5% over large-sample quantiles. Section VII-H checks the bound against PX4.

V. CROSS-LAYER EXPERIMENTAL SUBSTRATE

The evaluation runs on a cross-layer event schema released in the artifact [19], whose records join a network or environmental window to the navigation window it induces. For this paper we add a `perturbation_source` field (`net`, `gnss`, `fault`, `env`, `adv`), and every pairing carries a provenance flag: `measured_same_platform`, `physics_model` or `synthetic_alignment`. Adapters ingest the HCRL UAVCAN corpus [2], UAVIDS-2025 [1], a replay of the DATAMUt detector [34], the UAV Attack Dataset [3], [20] and UAV-EW-Bench [35]; an adapter for UAV-CAS [36] is included, but its release could not be retrieved for this study. Together they give 431,773 events (200,610 from real hardware), and the number of released `measured_same_platform` pairings is zero. Fault and environment sources follow the RflyMAD [37] and ALFA [38] taxonomies and the Dryden model [23]. The substrate supplies the constants of the network term (γ, s, L; Section II-B); the evasion study runs in simulation and is flagged `synthetic_alignment`. Third-party corpora are fetched from their origin under their own licences.

VI. EVASION STUDY DESIGN

A. Closed loop and perturbation models

The pilot models the lateral (cross-track) channel of a small multirotor cruising at $V = 8$ m/s through a 60 s mission segment. The airframe is a drag-damped double integrator, $\ddot{p} = u + c_d(w - \dot{p})$ with $c_d = 0.35\,\text{s}^{-1}$. It is driven by a proportional–derivative (PD) controller acting on the KF estimate and saturated at $4\,\text{m/s}^2$. The KF has states (position, velocity, accelerometer bias), propagates with the 20 Hz accelerometer, and corrects with 10 Hz GNSS fixes and the anchor fix every $T_a = 5$ s. Section D lists the simulator constants.

Environment. The lateral wind w follows the Dryden spectrum of MIL-F-8785C at 100 ft altitude. With W_{20} the wind speed at 20 ft, it has $\sigma_v = 0.1W_{20}/(0.177 + 0.000823h)^{0.4}$ and scale length $L_v = h/(0.177 + 0.000823h)^{1.2}$. The declared envelope is $W_{20} \in [15, 30]$ kt (light to moderate); severe turbulence (45–60 kt) lies outside it and serves as a negative control. Because the accelerometer measures the aerodynamic force, wind is consistent between the IMU and GNSS, as it is on a real airframe to first order.

Faults. Faults are drawn independently per mission: an accelerometer bias step (probability 0.3, $\sigma = 0.1\,\text{m/s}^2$); bias drift (0.2, with the random walk inflated tenfold); and actuator loss of effectiveness (LoE) of 10–30% (0.2).

GNSS. Noise has a white component ($\sigma = 0.5$ m) and a first-order Gauss–Markov drift ($\sigma = 0.8$ m, $\tau = 60$ s). Outages are exponential with mean 5 s (probability 0.3), and jamming-noise episodes triple σ for 10 s (probability 0.2).

B. Detectors

All four detectors observe only $O_{0:T}$. Each reduces a mission to one statistic S_D and alarms when $S_D > \theta_D$:

D1, signature monitor.
$S_1 = \max_k |z_k - z_{k-1}|$, the largest jump between consecutive fixes. It stands in for rule-based IDSs tuned to malicious signatures such as jumps and discontinuities.

D2, χ^2 innovation gate.
This is loosely modelled on innovation gating in PX4-class filters: it alarms when the normalised innovation squared (NIS) ν_k^2/S_k exceeds θ_2 on three consecutive fixes. (PX4's EKF2 rejects individual measurements rather than alarming.)

D3, SAVIOR-style CUSUM.
A model-based observer driven by the commanded acceleration with a no-wind model accumulates a cumulative-sum (CUSUM) statistic $g_k = \max(0, g_{k-1} + r_k^2 - (1+\kappa))$ on its GNSS residual [5].

D4, IMU–GNSS invariant.
A quadratic is fitted over 4 s to the residual between the GNSS position and the doubly integrated accelerometer; the detector alarms when the curvature, an acceleration discrepancy, exceeds θ_4 on five consecutive fixes. It is the cross-sensor analogue of the control invariants of Choi et al. [4].

Whenever C2′ holds, an anchor gate on the anchor fix's NIS (calibrated FPR 1%) is part of every stack. Thresholds are set on 4,000 benign missions so that each detector has the stated per-mission FPR; the combined stack ALL splits the budget equally among its four members.

C. Disguise families and the attacker

The offset $d(t)$, applied from an onset time t_0, belongs to one of five families:

- *step*: $d = A$ (the naive spoofer);
- *ramp*: $d = \min\{r(t-t_0), A\}$ (the "boiling frog");
- *fault mimic*: $\ddot{d} = \beta_0 + \beta(t)$, with β a random walk, capped at $|d| \le A$. It imitates accelerometer-bias drift, which the filter's bias state absorbs;
- *GNSS mimic*: $d = A|g(t)| + c(t-t_0)$, with g a unit Gauss–Markov process of attacker-chosen correlation time, imitating multipath or ionospheric drift;
- *weather mimic*: $d = A\,w(t-\ell)/\sigma_{\text{mod}} + b\min\{(t-t_0)/10, 1\}$, where σ_{mod} is the moderate-turbulence σ_v. The offset follows the true gust with a lag ℓ.

The attacker solves (3) with the cross-entropy method (CEM) [39]: 12 candidates per generation, 4 elites, 25 generations and 48 missions per candidate. Each (stack × family × operating point) combination gets its own optimiser seed in block 2xxx; stronger optimisers (CMA-ES [40], Bayesian optimisation) are the subject of the scaled attacker study (Section VII-E). Because a local optimiser can fail to find an evading parameterisation that exists, we do not read detector strength off the optimiser's own target. Instead, we *cross-evaluate* every optimised attacker against every stack.

D. Metrics and statistical protocol

For each optimised attacker, evaluated on fresh missions (block 4xxx), we report:

- the *evasion rate* (no alarm at any point in the mission);
- the attack's own contribution $\sup_t |e_{\text{adv}}|$ on evading missions, computed on the same realisation with the additive benign inputs set to zero but the parametric ones (outages, LoE) kept, as in Lemma 2;
- the Kolmogorov–Smirnov (KS) distance between each detector's statistic under attack and under benign conditions, a finite-sample estimate of the TV distance of Lemma 1 restricted to threshold tests.

In the cross-evaluated disguise matrix, a cell's contribution is the maximum over attackers that evade in at least 5% (and at least 10) of missions; we report the unfiltered maxima separately. The stealth budget $\delta_{\text{adv}}(\theta)$ is the supremum of the attack-only deviation over all evading missions, with no such filter. It is pooled over all 35 optimised attackers on calibration missions (block 1xxx). Every attacker uses the same missions at every operating point, and an attack that is stealthy against stricter thresholds is also stealthy against looser ones, so the pooled supremum is monotone in the operating point by construction.

For soundness, we re-optimise an independent attacker with fresh seeds (block 3xxx) for every family and FPR, against the full stack. We evaluate it on $n = 1{,}000$ fresh missions per operating point, crossing family, FPR, wind (light, moderate, envelope), margin m and θ_{net}. The theorem's event is $\{\text{fail} \wedge U\}$, so we count its occurrences k and compute the one-sided binomial p-value of H_0: $\Pr[\text{fail} \wedge U] \le \overline{p}(s_m)$. The p-values are adjusted across all operating points by Holm's procedure [41], and a violation is a Holm-adjusted $p < 0.05$. We also report the MCR conditional on U, with Clopper–Pearson intervals [42].

VII. Evaluation

The pilot comprises 1,909,600 simulated missions and ran in 29 minutes on two CPU cores. Every number below is generated from the released result files.

A. Calibration and detector operating points

At the 5% operating point, the per-detector FPR stays between 3.8% and 6.9% under moderate wind, and the anchor gate alone contributes 0.95%. One finding shapes the rest of the evaluation: wind and faults alone trip no detector (FPR 0.0% for every detector when either is the only benign source), whereas GNSS error alone produces up to 5.3%. The thresholds are therefore set by the tails of the GNSS error, and the tolerance they grant is the space an attacker exploits. Severe turbulence, outside the declared envelope, raises the FPR of the SAVIOR-style detector D3, which models no wind, to 8.2%; the others stay at or below 6.0%.

Figure 2 shows the operating curves against naive offsets. At 5% FPR the χ^2 gate detects a 5 m step in 100% of missions and the invariant D4 in 96%, but the signature monitor and D3 detect it in only 7% and 6%, which is close to chance. A 0.5 m/s ramp is caught by D2 (100%) and essentially by no other detector (≤5%).

B. RQ1: evasion and undetected deviation per detector

a) Cross-evaluation.: We evaluated each of the 35 optimised attackers against every stack. In 8 of 25 (stack, family) cells, the CEM run that targeted that stack found no parameterisation evading in at least 5% of missions, although an attacker optimised for a different stack does. Reading detector strength off the optimiser's own target would therefore have been wrong, so Table II reports the best attacker of each family, wherever it was optimised.

b) Evasion with the anchor gate armed.: With condition C2′ in force and its gate armed (Table II, top), every one of the five families evades every detector, and the full stack, in at least 87% of missions; for reference, benign missions raise no alarm in about 94% under the full stack. Each family has parameterisations that offset GNSS by about 1–2 m, and every

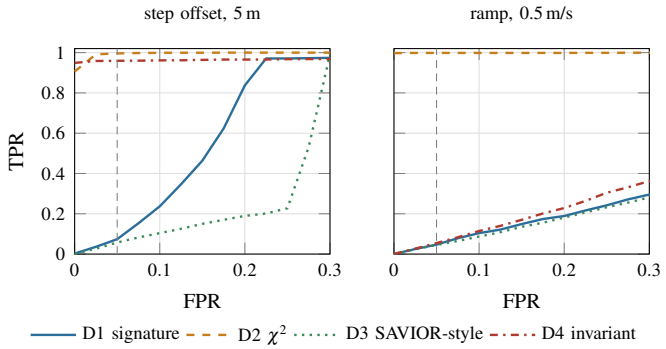


Fig. 2. Detector operating curves against two naive offsets, with thresholds swept over 4,000 benign missions (dashed line: the 5% operating point). Only D2 and D4 separate a naive 5 m step, and only D2 separates the ramp.

detector must tolerate offsets of that size to keep its FPR at 5%. Among attackers that evade in at least 5% of missions, the largest undetected contribution lies between 2.1 and 2.9 m, whichever detector is deployed. Rarer evasions go further: up to 5.6 m against D1, D3 (weather mimic). The pooled supremum against the full stack is 4.4 m (Fig. 3), and the stealth budget is $\delta_{\text{adv}} = 4.37$ m. Within that range, the trusted fix rather than the detector design determines how far an undetected attacker gets. The median KS distance between a detector's statistic under an evading attack and under benign conditions is 0.07, against 0.99 for detected ones. This is the empirical counterpart of Lemma 1; to the detector, an evading attack is almost indistinguishable from the benign population.

c) Evasion without the anchor gate.: The bottom of Table II repeats the cross-evaluation with the anchor still fused into the filter but its alarm removed, isolating what the detectors themselves discriminate. Evasion stays at or above 94%, but the undetected contribution now depends sharply on the pairing of disguise and detector.

- The fault mimic drives 73.9 m past D1, D3 and D4. Its slow, bias-shaped curvature has no jumps for D1, matches the bias drift that D4 tolerates by design, and looks like an unmodelled disturbance to D3.
- The GNSS mimic and the weather mimic exploit the two detectors that do not cross-check the IMU, D1 and D3 (up to 72.6 and 50.4 m). Among frequently evading attackers, D4 caps them at 3.3 m or below, because their acceleration content contradicts the accelerometer; a rare GNSS-mimic evasion still reached 19.2 m.
- Rare step offsets that happen to land in a benign noise trough evade D1, D3 and D4 in about 3% of missions and reach 26.8 m.
- Only the χ^2 gate D2, whose filter models the accelerometer bias, caps every family, at 9.9 m (fault mimic), including rare evasions (9.9 m); the full stack, which contains D2, is capped at 10.5 m.

This is the pattern adversarial equivalence predicts. The most effective disguise is the one the detector models as benign, namely bias drift for an invariant that tolerates bias, and drift

TABLE II

DISGUISE MATRIX AT 5% FPR, CROSS-EVALUATED OVER ALL 35 OPTIMISED ATTACKERS ON 1,000 HELD-OUT MISSIONS PER ATTACKER. *Top:* ANCHOR GATE ARMED; BEST EVASION RATE (%) / LARGEST UNDETECTED ATTACK CONTRIBUTION (M). *Bottom:* ANCHOR FUSED BUT ITS GATE REMOVED; LARGEST UNDETECTED CONTRIBUTION (M). CONTRIBUTIONS ARE TAKEN OVER ATTACKERS THAT EVADE IN AT LEAST 5% (AND AT LEAST 10) OF MISSIONS; RARER EVASIONS ARE REPORTED IN THE TEXT. BENIGN MISSIONS RAISE NO ALARM IN ABOUT 94%. DARKER CELLS ARE EVADED MORE OFTEN (TOP) OR ADMIT MORE DEVIATION (BOTTOM).

	step	**ramp**	**fault mimic**	**GNSS mimic**	**weather mimic**
anchor gate armed (C2′)					
D1 signature	89 / 2.1	91 / 2.4	94 / 2.6	92 / 2.5	88 / 2.9
D2 χ^2 gate	88 / 2.1	90 / 2.4	93 / 2.6	91 / 2.5	88 / 2.9
D3 SAVIOR-style	89 / 2.1	89 / 2.4	94 / 2.6	90 / 2.5	88 / 2.8
D4 IMU–GNSS inv.	87 / 2.1	90 / 2.4	92 / 2.6	91 / 2.5	89 / 2.9
ALL (stack)	88 / 2.1	90 / 2.4	94 / 2.6	91 / 2.5	90 / 2.9
anchor fused, gate removed					
D1 signature	2.2	2.4	73.9	72.6	50.4
D2 χ^2 gate	2.2	2.4	9.9	2.6	2.9
D3 SAVIOR-style	2.2	2.4	73.9	37.9	37.2
D4 IMU–GNSS inv.	2.2	2.4	73.9	2.6	3.3
ALL (stack)	2.2	2.4	10.5	2.6	2.9

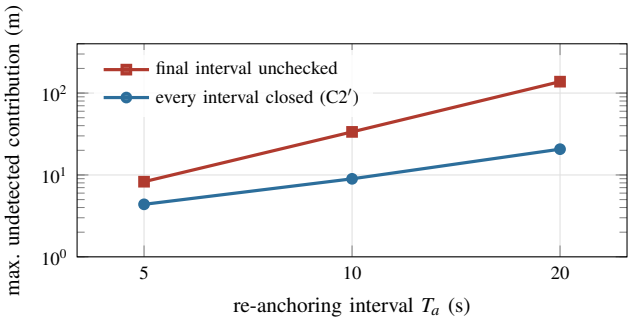


Fig. 3. Stealth budget against the full stack (5% FPR) versus the re-anchoring interval, pooled over 30 attackers (15 tuned per T_a and the 15 optimised against the full stack) and maximised over the five disguise families, on the same missions at every T_a. With every interval closed, the largest undetected contribution grows from 4.4 to 20.6 m (4.7×). With the final segment before mission end unchecked, it grows 17×, from 8.3 to 137.8 m.

or wind-shaped motion for detectors without an inertial cross-check.

d) Re-anchoring interval and terminal window.: Figure 3 sweeps the re-anchoring interval against the full stack, pooling all attackers on common missions. With every interval closed, the largest undetected contribution grows from 4.4 m at $T_a = 5$ s to 20.6 m at 20 s (4.7×). The attacker must return near zero before each trusted fix, so the budget is set by how far it can travel and come back within one interval. When the segment before mission end is left unchecked, no return leg is needed, and the contribution reaches 137.8 m, 7× the closed value. An equivalence attacker therefore spends its budget in the terminal window.

C. RQ3: decomposition of the bound's looseness

Figure 4 decomposes the tail of the benign budget; the ratios below are at $s = 4$ m, and their ordering changes in the far tail,

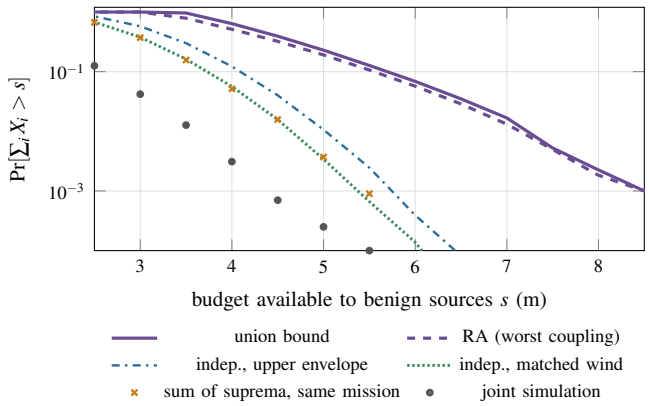


Fig. 4. Decomposition of the looseness (tail of the benign budget). Union and RA nearly coincide. The steps down from RA are: to independent sums at the upper wind envelope (dependence gap); to independent sums at the wind actually flown (envelope gap); to the sum of per-source suprema on the same missions, which confirms that the simulated sources are independent; and to the joint simulation (temporal gap: sources seldom peak at the same instant).

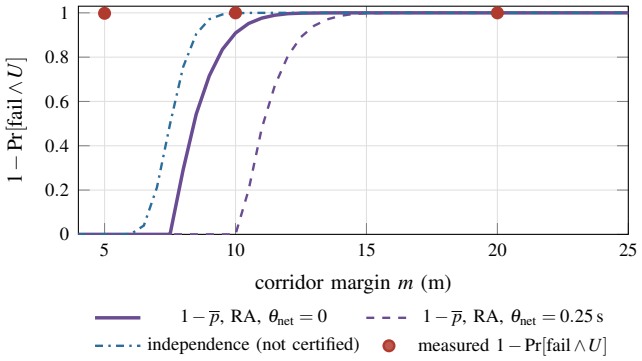


Fig. 5. Certified bound versus measurement at 5% FPR. The measured points are the worst case over five held-out attackers and three wind conditions. The bound stays below every measurement. Including the network term shifts it right by $e^{LT_c}\delta_{\text{net}} = 2.65$ m.

where few joint exceedances remain.

1) For two sources, the union and RA bounds agree to within 0.001, as Makarov's theorem requires. For three, the RA tightens the union bound by 20% (0.6390 to 0.5133). Under unknown dependence, the union bound is nearly the right quantity.
2) The worst coupling exceeds independent sums at the same marginals by 4× (*dependence gap*). Evaluating the wind at the upper envelope rather than at the conditions actually flown costs a further 2.2× (*envelope gap*).
3) Independent sums at matched wind agree with the sum of per-source suprema measured on the same missions to within 1.09×, consistent with the benign sources being independent, as they are by construction in the simulator.
4) The sum of per-source suprema exceeds the supremum of the joint deviation by 16.6× (*temporal gap*; 62 joint exceedances in 20,000 missions). Sources seldom peak at the same instant, but a budget built from per-source suprema must assume they do.

The dependence gap can be closed only by knowing the coupling, which an attacker who controls jamming denies the defender. The envelope gap shrinks with better wind knowledge. The temporal gap, the largest of the three at this s, can be closed without either, by bounding the supremum of the summed trajectory directly, for example with a reachable-set computation per certification window. Condition (C5), which licenses adding per-source suprema, held in 99.63% of joint missions.

D. RQ2 and RQ4: soundness of the bound under evasion

Pooling all attackers on calibration missions gives a stealth budget of $\delta_{\text{adv}} = 4.37$ m, attained by the fault mimic. The budget is identical at 1%, 5% and 10% FPR. It is a single pooled maximum over common missions, so this says that no attacker in the pool gains from looser thresholds in that range, not that thresholds never matter.

Table III reports, for each operating point, the bounds on $\Pr[\text{fail} \wedge U]$ and the largest observed frequency of that event over five held-out attackers and three wind conditions. Across 270 distinct operating points and 30,541 undetected-attack missions, no point shows a Holm-significant violation (0 of 270); the scaled test of Section VII-E overturns this. Wherever the bound is informative ($\overline{p} < 1$), the observed frequency of undetected failure was zero. The bound is therefore conservative, and the zero violations alone carry little evidence; the negative controls below show that the test can fail.

Three further checks qualify this result.

- *The empirical budget is not a proven bound.* The held-out attacker exceeded δ_{adv} in 1 of 45 (family × FPR × wind) cells, reaching 5.13 m with the fault mimic at 5% FPR (1.17×). The bound absorbed the excess, which led us to strengthen the optimisers and derive an analytic budget (Sections VII-E and VII-F). In 6 cells, the held-out optimiser found no evading attack at all.
- *Lemma 2 holds where (C1) holds.* Attack/benign superposition held in at least 94.4% of missions at every point. A control run with the actuator limit removed (5% FPR, envelope wind, all five families) restored it in 100.00% of missions (largest excess 7.8e-15 m). We attribute the larger excesses at other points, up to 11.03 m, to the same mechanism, saturation, which (C1) excludes; these points were not re-simulated.
- *Independence is not certified.* The independence bound $p_{\perp}$ was not violated either, because the simulated sources are independent. It would claim a 0.9 floor at $m = 8.5$ m, against 10.0 m for the certified bound (Fig. 5); nothing guarantees independence when an attacker controls jamming.

a) Negative controls and the power of the test.: A soundness test that cannot fail is uninformative, so we broke two conditions deliberately and re-optimised the attacker under each (Table IV).

- *(C2′) broken, no re-anchoring.* Undetected deviation reaches

TABLE III

SOUNDNESS OF THEOREM 1 ON HELD-OUT ATTACKERS. BOUNDS ON $\Pr[\text{FAIL} \wedge U]$: UNION p_U, RA $\overline{p}$ (CERTIFIED) AND INDEPENDENCE $p_\perp$ (NOT CERTIFIED). OBS. = LARGEST OBSERVED FREQUENCY OF $\{\text{FAIL} \wedge U\}$ OVER FIVE FAMILIES AND THREE WIND CONDITIONS; MCR_U = SMALLEST COMPLETION RATE AMONG UNDETECTED MISSIONS; VIOL. = HOLM-SIGNIFICANT VIOLATIONS OF $\overline{p}$ AND OF $p_\perp$. $\delta_{\text{adv}} = 4.37$ M. ROWS WITH $\theta_{\text{net}} = 0.25$ S SHIFT THE MARGIN ANALYTICALLY BY $e^{LT_c}\delta_{\text{net}} = 2.65$ M; THE NETWORK TERM ITSELF IS NOT SIMULATED. p_U IS A PROVEN UPPER BOUND; THE RA COLUMN IS A NUMERICAL APPROXIMATION OF THE SHARP $\overline{p}$, REPORTED AS $\min(\text{RA}, p_U)$.

FPR	θ_{net}	m	p_U	$\overline{p}$	$p_\perp$	obs.	MCR_U	viol.	
1%	0.00	5	1.000	1.000	1.0000	0.0030	0.996	0	0
1%	0.00	10	0.109	0.091	0.0016	0.0000	1.000	0	0
1%	0.00	20	0.000	0.000	0.0000	0.0000	1.000	0	0
1%	0.25	5	1.000	1.000	1.0000	0.4640	0.432	0	0
1%	0.25	10	1.000	1.000	0.5878	0.0000	1.000	0	0
1%	0.25	20	0.000	0.000	0.0000	0.0000	1.000	0	0
5%	0.00	5	1.000	1.000	1.0000	0.0020	0.998	0	0
5%	0.00	10	0.109	0.091	0.0016	0.0000	1.000	0	0
5%	0.00	20	0.000	0.000	0.0000	0.0000	1.000	0	0
5%	0.25	5	1.000	1.000	1.0000	0.4250	0.508	0	0
5%	0.25	10	1.000	1.000	0.5878	0.0000	1.000	0	0
5%	0.25	20	0.000	0.000	0.0000	0.0000	1.000	0	0
10%	0.00	5	1.000	1.000	1.0000	0.0010	0.999	0	0
10%	0.00	10	0.109	0.091	0.0016	0.0000	1.000	0	0
10%	0.00	20	0.000	0.000	0.0000	0.0000	1.000	0	0
10%	0.25	5	1.000	1.000	1.0000	0.3880	0.467	0	0
10%	0.25	10	1.000	1.000	0.5878	0.0000	1.000	0	0
10%	0.25	20	0.000	0.000	0.0000	0.0000	1.000	0	0

TABLE IV

NEGATIVE CONTROLS AT 5% FPR. UNDET. = UNDETECTED MISSIONS OUT OF 1,000; FAILS = FAILURES AMONG THEM; VIOL. = HOLM-SIGNIFICANT VIOLATION OF $\overline{p}$; DEV. = LARGEST UNDETECTED DEVIATION (M).

Control	disguise	m	undet.	fails	$\overline{p}$	viol.	dev.
C2′: no anchor	ramp	5	907	907	1.000	no	47.8
C2′: no anchor	ramp	10	907	907	0.091	**yes**	47.8
C2′: no anchor	ramp	20	907	907	0.000	**yes**	47.8
C2′: no anchor	fault	5	729	719	1.000	no	308.5
C2′: no anchor	fault	10	729	683	0.091	**yes**	308.5
C2′: no anchor	fault	20	729	361	0.000	**yes**	308.5
C4: severe wind	fault	5	889	106	1.000	no	96.7
C4: severe wind	fault	10	889	37	0.091	no	96.7
C4: severe wind	fault	20	889	14	0.000	**yes**	96.7
C4: severe wind	weather	5	0	0	1.000	no	0.0
C4: severe wind	weather	10	0	0	0.091	no	0.0
C4: severe wind	weather	20	0	0	0.000	no	0.0

308.5 m, and the test flags 4 of 6 points (fault and ramp), every one where the bound is informative.

- *(C4) broken, severe turbulence (45–60 kt).* The fault mimic reaches 96.7 m undetected, and the test flags the bound at $m = 20$ m, where $\overline{p} = 0$ because the empirical marginals have finite support. At $m = 10$ m the observed frequency stays below $\overline{p}$, so this break is detected only where the bound is tightest. The weather mimic was detected in every mission, leaving no undetected missions to test.

The test thus detects unsoundness when a stated condition fails and enough undetected missions remain to measure it, which is the behaviour the audit loop of Fig. 1 relies on.

E. RQ5: a scaled attacker study and soundness test

The pilot's zero violations rested on five CEM attackers per cell and a conservative bound. We therefore strengthened the attacker and the test before drawing conclusions, in a scaled attacker study. 126 further attackers were optimised against the full stack: CMA-ES (population 16, 200 generations, one restart) [40], Bayesian optimisation with a tree-structured Parzen estimator (400 trials), and the same TPE with a tail objective (the 90th percentile of the undetected deviation instead of its mean). Every candidate was scored on 128 missions with common random numbers, and a sixth *mixed* family switches between any two disguises mid-mission. The scaled soundness test re-optimised 5 independent CEM attackers per (family, FPR) cell and evaluated each on 5,000 fresh missions under three wind conditions: 1,620 tests over 797,435 undetected-attack missions.

a) Growth of the stealth budget.: The pooled budget rose from 4.37 to 5.46 m at 1% and 5% FPR (1.25×) and to 5.04 m at 10%, attained by the gnss mimic (tail-objective TPE). Only the tail objective found more than the pilot's CEM: the mean-objective CMA-ES and TPE peaked at 3.85 and 3.09 m, so an attacker that optimises its typical deviation does not set the empirical supremum. Even the larger budget was exceeded by a held-out attacker in 12 of 270 cells, by the mixed family up to 7.04 m. An empirical supremum is a lower bound on δ_{adv}, and further search continued to raise it.

b) Holm-significant violations and their diagnosis.: With $\delta_{\text{adv}} = 5.46$ m, the test flagged 5 of 1,620 points, all at $m = 20$ m, where the bound claimed $\overline{p} = 0$. We re-simulated each flagged cell with its seeds (`r1_violation_diagnosis.csv`). The violations come from 3 missions, and the attacker caused none of them. Each fails under its benign inputs alone (benign-only deviation 19.2–34.5 m), the attack adds at most 2.3 m, and all three saturate the actuator with or without the attack. Two conditions of Theorem 1 failed, neither of which concerns disguise.

- *(C3) for the marginals.* The empirical marginal of each source has finite support. Beyond the largest of 4,000 calibration samples, $\overline{p}$ is zero, and the bound claims a certainty that no finite sample can justify. One benign mission in 5,000 fails at $m = 20$ m.
- *(C1), which was not enforced.* Superposition held in at least 90.9% of missions, and every violating mission saturated. The pilot had stated (C1) as an assumption, but no component of the stack checked it at run time.

This is a negative result about the pilot instantiation rather than about Theorem 1. The theorem's conclusion requires its conditions, and the scaled test found two that the instantiation did not secure. We report the violations as found. The revision below fixes the instantiation before any new data are examined and does not tune the old run until it passes.

F. A revised instantiation evaluated on fresh data

We changed three things, fixed them before generating any new data, and evaluated them on a new seed block. A smoke test of the code used a separate block, which we did not reuse.

1) *A saturation alarm D_{sat}.* The autopilot raises an alarm at any epoch at which the commanded acceleration reaches its limit. This turns (C1) from an assumption into a run-time check; on the event U the loop never saturated, so Lemma 2 applies exactly. The alarm costs benign missions 1.1% in false alarms (5% operating point, envelope wind). The full stack's false-alarm rate rises from 4.8% to 5.6%.
2) *A finite-sample p-box for (C3).* Each marginal tail is replaced by its Clopper–Pearson upper bound, taken simultaneously over the split grid (Bonferroni over 3×501 grid points, $\alpha = 0.05$), and aggregated by the union bound. With probability at least 0.95 over the calibration samples, $\overline{p}$ is then a valid upper bound everywhere on the grid. It never reaches zero, and with 4,000 samples per source it levels off at $1 - 0.992$.
3) *Two stealth budgets, tested separately.* The first is the calibration supremum of the scaled attacker study, $\delta_{\text{adv}} = 5.46\,\text{m}$. It was not widened to the 7.04 m that the study's held-out attacker reached, because a constant widened after looking at held-out data would make the new test circular. The second is the analytic bound below, which holds for every attack.

a) An analytic stealth budget.: With noise removed, the linear loop maps the attack offsets to the attack-only deviation linearly, so the worst undetected deviation is a linear programme over the offsets, constrained by D1, the anchor gate and D_{sat}, with benign margins at their $1 - \varepsilon/3$ calibration quantiles ($\varepsilon = 0.01$), the worst LoE in $[0.7, 1]$ and a free onset (`r8_analytic_bound.csv`). Section IV-E states the attacker class and the result. At $T_a = 5\,\text{s}$ the budget is 31.9 m (33.5 m with the finite-sample correction), 5.8× the calibrated δ_{adv} and 4.5× the largest deviation any optimised attacker reached (7.04 m); the gap measures what D2–D4 contribute beyond what has been proved.

b) Result on fresh seeds.: We re-optimised 72 held-out attackers against the revised stack: three CEM seeds per (family, FPR) cell plus one tail-objective TPE. Each was evaluated on 5,000 missions under three wind conditions, from a seed block never used before (510,848 undetected-attack missions).

- *No violation under either budget.* With the empirical budget, the test found 0 Holm-significant violations in 1,296 tests; with the LP budget, 0 in 1,296. Superposition was exact on every undetected mission (largest excess 0.0e+00 m). Missions like the benign-tail failures that broke the pilot instantiation saturate, so they now raise the alarm and leave the event U.
- *The empirical budget was exceeded again.* In 6 of 216 cells a held-out attacker went beyond $\delta_{\text{adv}} = 5.46\,\text{m}$, up to 6.25 m (weather mimic, CEM). Benign slack absorbed it and no violation followed, but the usable stealth budget is still not a sound constant. The LP budget was never exceeded; it can be exceeded only with probability ε, or on missions with a GNSS outage, which its proof does not cover.
- *What the bound certifies.* With the empirical budget, the floor at 5% FPR is 0.541 at $m = 10\,\text{m}$ and 0.992 at $m = 20\,\text{m}$. With the LP budget it is 0.000 at both, so it certifies nothing in the corridors we test.
- *Limited power.* Of the 648 tests at which the bound is informative, 0 saw any undetected failure. The revision is therefore not refuted, nor is it confirmed to be tight. The negative controls remain the evidence that the test can fail, and the scaled run of Section VII-E shows that it does when a condition is missing.

G. RQ6: effect of a measured GNSS marginal

The pilot's GNSS marginal is simulated (99th percentile 2.8 m). The artifact contains measured navigation error from the real GPS-jamming flight of the UAV Attack Dataset [20]: 496 one-second samples over a noise episode and a loss of fix, whose error grows with staleness to 6.2 m at 10 s and 7.0 m at 15 s (spoofing samples belong to δ_{adv} and are excluded). We replaced each simulated degradation episode by the larger of its simulated deviation and the measured envelope at the episode's staleness, and in a stress variant set every episode to the measured 45 s duration. This never makes a mission's GNSS deviation smaller, but it assumes, without measuring it, that navigation error becomes cross-track deviation one for one.

At $m = 10\,\text{m}$ and 5% FPR, with $\delta_{\text{adv}} = 5.46\,\text{m}$, the floor estimated from the empirical marginals falls from 0.69 (simulated GNSS) to 0.34 (measured envelope) and 0.32 (45 s episodes). With the finite-sample band of Section VII-F, the values are 0.54, 0.21 and 0.19. The bound is only as good as its GNSS marginal; a simulated marginal overstated the certified floor by a factor of two to three at this margin. It is the jamming-induced tail, rather than the attack, that the corridor must absorb. At $m = 20\,\text{m}$ every variant certifies 0.992, the ceiling that 4,000 calibration samples per source permit.

H. RQ7: transfer to PX4

a) Harness.: We repeated the key experiments in PX4 v1.14.3 SITL with its SIH physics, flying a 60 s straight segment at about 8 m/s in mission mode, 12,321 missions in all. The GNSS offset and the benign envelope (gust, GNSS noise, outages, jamming, IMU faults, LoE) are injected inside the simulator from seeded streams, so each attacked mission has a twin with the same noise and the attack off; the attack-only deviation is the largest difference between the two. Stock EKF2 v1.14 fuses vision position only relative to a bias estimate while GNSS is active, so a trusted fix would be absorbed; a small EKF2 change fuses the fix every 5 s as an absolute position. D1–D4 and the anchor gate are computed offline from the flight logs, with D2 taken from EKF2's own GNSS innovation test, and thresholds are recalibrated on 465 benign missions. The protocol was pre-registered in the artifact before the first attacked flight. Two dated deviations followed. In 0%–7% of missions a start-up race in the harness sent PX4 into a failsafe before the window, independently of the attack, and these missions are excluded. The PX4-optimised attacker of each family was re-selected after this exclusion. Flying one benign mission twice differs by 0.67 m in the median and 1.53 m at most, a floor under every attack-only number below.

TABLE V
PX4 SITL, 5 S RE-ANCHORING, 5% FPR. DEVIATIONS (M) ARE ATTACK-ONLY ON UNDETECTED MISSIONS. RO: REDUCED-ORDER STUDY (DETECTION: D2).

	PX4	RO
Step 5 m caught: EKF2 / stack+gate	97.2% / 100%	100%
Ramp 0.5 m/s caught: EKF2 / stack+gate	99.7% / 100%	100%
Transferred attackers: evasion	78%–88%	69%–87%
Transferred attackers: largest deviation	2.2–4.8	1.5–2.9
Worst of 35, stack+gate	4.75	4.4
Worst of 35, no gate: stack / D4 / D1	11.6/29.9/48.0	
PX4-optimised: evasion; largest dev.	80%–88%; 2.8	
Stealth budget δ_{adv}	4.75	4.37/5.46
No anchor, ramp: evasion; drift	93%; 17.5	
Re-anchoring 1 s: worst, stack+gate / D1+gate	2.44/3.16	bound 10.6

b) Results.: Table V summarises the study. Against naive offsets, EKF2's own gate caught the step in 97.2% and the ramp in 99.7% of missions, and the full stack with the anchor gate caught both every time. The 35 attackers optimised in the reduced-order model, flown without re-optimisation, still evaded the full PX4 stack: the five primary ones in 78%–88% of missions, but their median attack-only deviation was only 1.4–2.1 m, near the replay floor, and the worst over all attackers was 4.75 m. Attackers optimised against PX4 itself evaded in 80%–88% of missions with at most 2.8 m; no GNSS or weather mimic evaded at all. The PX4 stealth budget, 4.75 m (median 1.5, 95th percentile 2.2 m), lies between the reduced-order 4.37 and 5.46 m. Without the trusted fix, a 0.5 m/s ramp evaded in 93% of missions and moved the vehicle by the full 17.5 m, as the finiteness result predicts. For the stack of the proved bound (D1 and the anchor gate), the worst PX4 case was 7.1 m, inside the 33.5 m bound at 5 s; this is an empirical consistency check, since the proof covers the linear loop only. D3's threshold, set by benign outages, made it as likely to fire on a naive attack as on a benign mission (7% and 5%). With re-anchoring every 1 s (pre-registered with two predictions), both held: the worst undetected deviation was 3.16 m for the bound's stack, inside the 10.6 m bound and below the 5 s value, and 2.44 m for the full stack with the gate (2.67 m even without it); the naive ramp evaded in 0% of missions.

c) The pre-registered reading did not hold.: The protocol fixed two conditions for a qualitative transfer of the disguise matrix. (a) With the gate armed, every primary attacker should evade in at least half of the missions with at most twice its reduced-order deviation; 4 of five families met it, but the step reached 4.75 m against a limit of 4.22 m. (b) With the gate removed, EKF2's gate should cap every family at 6.77 m while D4 does not; D4 admitted 29.9 m to the fault mimic, but EKF2's gate also admitted 11.6 m. What transferred is the ordering: the anchor gate decides how far an undetected attacker gets, the inertial invariant D4 is bypassed by the fault mimic, and without the gate the damage is two to three times larger in PX4 than in the reduced-order model.

VIII. DISCUSSION

a) Summary of findings.: In the reduced-order study every disguise evades every detector, and only the anchor gate caps the undetected deviation for all of them (RQ1); the repaired bound survived fresh attackers (RQ2–RQ5); a measured GNSS marginal cut the floor two- to threefold (RQ6); and in PX4 the stealth budget transferred while the full disguise matrix did not, and 1 s re-anchoring held every undetected deviation within 3.16 m (RQ7).

b) Design rules for defenders.: Five design rules follow from the measurements, and each mitigates the attack described in this paper.

1) *Re-anchor, and size the interval rather than the threshold.* With a trusted fix every 5 s and its gate armed, frequent evasions contributed at most 2.9 m and the pooled supremum was 4.4 m, for every detector and disguise. Without the gate, the contribution reached 73.9 m. Lengthening the interval from 5 to 20 s multiplied the budget by 4.7 (Fig. 3), while moving the operating point from 1% to 10% FPR left the pooled budget unchanged in the pilot (Section VII); the proved budget falls almost linearly with T_a (Table I); in PX4, 1 s instead of 5 s cut the worst undetected deviation from 4.75 to 2.44 m. For a corridor of margin m, choose T_a such that $\delta_{\text{adv}}(T_a) \le m - e^{LT_c}\delta_{\text{net}} - s^\star$, where $s^\star$ is the smallest budget with $\overline{p}(s^\star)$ below the tolerated risk.
2) *Close the last interval.* An unchecked segment before mission end lets the attacker skip the return leg and multiplied the budget by 7 at $T_a = 20$ s. A trusted fix therefore matters most during landing and terminal approach.
3) *Model the inertial bias as well as cross-checking the inertial sensor.* Without the anchor gate, the only detector that capped every disguise was the one whose filter models the accelerometer bias (D2). D1 and D3 admitted tens of metres to the GNSS and weather mimics, and even the inertial invariant D4 admitted 73.9 m to the fault mimic, whose bias-shaped curvature it tolerates by design.
4) *Certify the undetected outcome rather than the alarm.* Theorem 1 bounds $\Pr[\text{fail} \wedge U]$ for every attacker whose undetected contribution stays within δ_{adv}, without relying on the detector catching any particular disguise. A floor computed from benign marginals alone and gated on silence has no such property (Proposition 1).
5) *Alarm on what the certificate assumes.* The pilot assumed an unsaturated loop, and the scaled test failed exactly where the loop saturated. A saturation alarm turns the assumption into a run-time check, at a cost of 1.1% benign false alarms. Every condition of a deployed certificate should be either proved or monitored.

c) Limitations.:

- *Reduced-order model.* The proof and the soundness test cover one lateral channel with a linear KF. The PX4 study (Section VII-H) ran in SITL, needed a small EKF2 change, and transferred the stealth budget but not the full disguise matrix; a model-mismatch term to EKF2 is not yet bounded.

- *Empirical adversarial term.* The usable δ_{adv} is a pooled supremum over optimised attackers; stronger optimisers raised it by $1.25\times$, and a held-out mixed-family attacker exceeded it (7.04 m). The proved budget (Section IV-E) omits the run-based detectors D2 and D4, which need mixed-integer constraints; a white-box gradient attacker is left for future work.
- *Finite calibration.* With 4,000 samples per source, the simultaneous band certifies at most 0.992; a floor of $1-10^{-k}$ needs several times 10^{k+1} missions per source.
- *Limited power.* Where the revised bound is informative, no undetected failure occurred, so the bound has not been stressed near its value, although the test has shown that it can fail (Section VII-E).
- *GNSS marginal.* Only the GNSS marginal was replaced by a measured one (Section VII-G), from one campaign on one airframe; the fault marginal remains simulated.
- *Analytic network term.* δ_{net} enters as the certified tube of Section II-B rather than through simulation; injecting measured relay staleness into SITL is a stated extension.
- *Covert attackers only.* The bound covers undetected failure. An attacker indifferent to detection is bounded instead by detection latency and the failsafe, which we do not certify here.
- *No IMU injection.* An attacker who can also inject into the IMU (acoustically [27], [28] or through bus reconfiguration [43]) defeats the inertial cross-checks, and condition C2′ alone then limits the attacker.

IX. Related Work

a) Stealthy and disguised attacks on control systems.: Stealthy attacks have been studied from false-data injection against χ^2-monitored Kalman filters [44], [45] and the attack space of Teixeira et al. [46], through impact limited by a false-alarm budget [13] and reachable-set metrics [12], [47], to ε-stealthiness in KL divergence [15], attacks that alarm at the false-alarm rate [16], undetectable unknown inputs [48], interval observers under bounded noise [49] and secure estimation under sparse attacks [50]. Each treats the benign perturbation as a single, known nuisance. Fault-mimicking actuator attacks [9] and separability analyses [51], [52] concern a fixed attack against a fixed detector; equivalence concerns the attacker's best response, and against every detector we built some small-amplitude disguise evaded (Section VII), a question power grids share [53]–[56].

b) Attacks on UAV navigation.: GNSS spoofing has advanced from portable spoofers and takeover conditions [24], [25] to covert UAV capture [26], adaptive hijacking [57] and attacks on multi-sensor fusion [58], [59], surveyed in [60], [61]. Stealthy attacks against control-based defences [6], [62], black-box GPS attacks [10], [63], optimal attacks on nonlinear estimators [64], sensor reconfiguration [43], wind-exploiting spoofers [11], real-time redirection [65], acoustic inertial injection [27], [28] and the hardness of physical sensor attacks [66] establish that stealth is possible, but none treats the disguise as the attacker's optimisation variable or links undetected deviation to a mission-level certificate.

c) Detection and recovery.: Control and physical invariants [4], [5], software sensors [67], PID-Piper [68], diagnosis-guided recovery [69], safety-critical variable monitoring [70], PWM-fused IDSs [71] and inertial-attack recovery [72] are the defences an attacker in our model must evade; active exposure [73] and fuzzing [74], [75] complement them. Hardening learned intrusion detectors against evasion [76] or calibrating their false-alarm operating point [77] does not bound what an undetected attacker achieves. Power-distortion and INS-aided GNSS monitors with integrity-risk bounds [78], [79] are the closest analogue of our budget, without a fault or environment term or unknown dependence.

d) Certificates and reachability.: Randomized smoothing [7], [8], [80] and its reinforcement-learning extensions [81]–[83] certify models or rewards, not missions. Set-based reachability [84]–[86], tube model predictive control [87], funnels [88], [89], control barrier functions [90], Hamilton–Jacobi reachability [91] and Simplex [14] supply envelopes, and probabilistic zonotopes assume independent components [85], [92]. The nearest prior work to our budget is stochastic reach-avoid games against an adversary plus nature [93], with a known kernel, and Wasserstein distributionally robust reachability for stealthy injection under ambiguous noise [94], without dependence between components or a mission-level floor; conformal methods [95], [96] assume exchangeability, which an adaptive attacker breaks.

e) Dependence uncertainty.: Sharp bounds on sums with fixed marginals [17], [18], [30], the Rearrangement Algorithm [31]–[33], [97], [98] and probability-bounds analysis [22], [99]–[101] are mature in finance, reliability [102] and power systems [103]–[105], and appear in verification of neural networks [106]–[108] and motion planning [109]. To our knowledge, no prior work applies them across an adversarial, a stochastic and an environmental perturbation to certify a cyber-physical mission outcome.

X. Conclusion

A UAV's navigation stack cannot distinguish an attacker from the weather, and an attacker who knows this will exploit it. We turned that observation into a threat model, ε-observational equivalence, and a bound on undetected failure for every coupling of the benign sources. Optimised disguises evaded every detector, and for an explicit adaptive GNSS attacker the undetected deviation is now proved on the linear loop: finite only with a trusted re-anchoring fix and 10.6 m at 1 s, while PX4 SITL flights at that interval stayed within 3.16 m. The re-anchoring interval, not a threshold, is therefore the defender's design parameter. Engineers can size it to the corridor a mission must hold; certification authorities can require a bound on undetected failure, with every condition proved or monitored, not only a detection rate; and researchers gain a way to state and falsify such bounds under pre-registered protocols. Bounding the mismatch to EKF2 and proving the run-based detectors come next.

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Ethics Considerations

This work studies how an attacker can disguise interference with a UAV's navigation as a benign disturbance. The method is dual-use, so we weighed its benefits and harms before running any experiment, following the ethics guidance of the call for papers.

a) Stakeholders and harms.: The stakeholders are UAV operators, people and property under flight paths, autopilot maintainers and regulators. The potential harm is that an adversary uses the disguise taxonomy, or the optimiser, to make a real spoofing attack harder to detect.

b) Assessment of benefits and risks.:

- *The attack capability is not new.* Detector-agnostic stealthy GPS attacks against UAVs are already published [10], as are learned spoofers that exploit wind compensation [11]. Our contribution is the measurement of which disguise defeats which detector, together with a defence that does not depend on detection.
- *Each attack family is accompanied by a mitigation.* The mitigations are re-anchoring-interval sizing, detector diversity across disguise classes, and the source-agnostic floor.
- *No experiment emitted radio-frequency energy.* All attacks act through simulator interfaces, and the released harness contains no transmit-capable code path.
- *No human subjects or personal data are involved.* The third-party corpora are public, are used under their licences, and are not redistributed.

c) Disclosure.: The reduced-order study uses a generic Kalman filter. The PX4 SITL study (Section VII-H) shows that small disguised GNSS offsets pass EKF2's innovation gating and that, without a trusted fix, a slow ramp moves the vehicle by the full offset undetected. This follows from tolerating benign GNSS error rather than from a software defect, and the mitigation is a trusted re-anchoring fix. We will share the results with the PX4 and ArduPilot security contacts at least 90 days before publication.

Open Science

We release, through an anonymised repository for review, https://anonymous.4open.science/r/Weather-Certified-UAV-Autonomy/, and a permanent DOI at camera-ready [19]:

- the schema (versions 0.1 and 0.2) and its adapters;
- the pairing pipeline, with mandatory provenance flags;
- the dependence-bound library (union bound, Rearrangement Algorithm, independence, comonotone);
- the simulator, the four detectors and the evasion harness, restricted to simulator interfaces;
- the PX4 v1.14.3 SITL patch (GNSS offset injection, benign envelope, trusted anchor), the fleet driver, the offline detectors and the pre-registration with its dated deviations;
- every result file behind every number in this paper;
- the scripts that regenerate all macros, figures and tables from those files.

Seeds are partitioned by role (calibration, optimisation, held-out attacker, held-out evaluation, negative controls). A CPU-only path reproduces every figure in about 29 minutes on two cores. Third-party corpora are fetched from their origin under

their own licences. We target the Available, Functional and Reproduced badges.

Appendix A
Notation and Acronyms

Symbol	Meaning
$x, \hat{x}$	true and estimated navigation state
$e(t)$	cross-track deviation from the nominal path (m)
m	corridor margin (m); completion iff $\sup_t \lvert e\rvert \le m$
$L,\ T_c$	closed-loop Lipschitz constant; certification horizon (1 s)
$\theta,\ \theta_D$	detector operating point; threshold of detector D
$\Delta(\theta),\ s,\ H$	hidden staleness; contact-window slack; malicious hops
γ	delay-to-position rate (m/s), campaign supremum
$\delta_{\text{net}}, \delta_{\text{adv}}$	deterministic network and adversarial-offset budgets (m)
$\delta_{\text{gnss}}, \delta_{\text{fault}}, \delta_{\text{env}}$	stochastic and bounded-unknown budget terms
X_i	sup-deviation of the closed loop under source i alone
S_i	survival function of X_i, $S_i(t) = \Pr[X_i > t]$
$\underline{F}, \overline{F}$	lower and upper CDFs of a p-box ($\underline{F}$ is stochastically largest)
$\overline{p},\ p_U$	worst-case (all couplings) and union (standard) tail bounds
$f(m)$	certified floor, $1-\overline{p}(s_m)$
U	event that the detector stack raises no alarm
T_a	re-anchoring interval (C2′)
$O_{0:T}$	defender's observations over the mission
$P_a,\ P_d$	laws of $O_{0:T}$ under attack a / disguise d
$\mathscr{D}_k$	disguise family k

Acronym	Expansion
CDF	cumulative distribution function
CEM	cross-entropy method
CMA-ES	covariance matrix adaptation evolution strategy
CUSUM	cumulative sum (sequential change detection)
EKF / KF	extended Kalman filter / Kalman filter
FPR	false-positive rate
GNSS	global navigation satellite system
IDS	intrusion detection system
IMU	inertial measurement unit
KL / KS	Kullback–Leibler divergence / Kolmogorov–Smirnov distance
LoE	loss of effectiveness (actuator)
MCR	mission-completion rate
NIS	normalised innovation squared
PD	proportional–derivative (controller)
p-box	probability box
RA	Rearrangement Algorithm
SITL	software in the loop
TV	total-variation distance
UAV	unmanned aerial vehicle

Appendix B
Proof Details

a) Proposition 2(b): sharpness for two sources.: For $d = 2$, Makarov [17] showed that $\sup_C \Pr_C[X_1 + X_2 > s] = \min\{1, \inf_t (S_1(t^-) + S_2((s-t)^-))\}$, which equals $p_U(s)$ at every continuity point of the empirical survival functions. The extremal coupling makes the events $\{X_1 > t^\star\}$ and $\{X_2 > s - t^\star\}$ disjoint, by pairing the upper tail of one source with the body of the other [30]. An attacker who controls when a stochastic source fires (for example, the timing of jamming-induced outages relative to gusts) can move the coupling towards this extremal one.

b) Proposition 2(d): comonotone and independent gaps.: Let $X_1 = \cdots = X_d = X$ almost surely, with survival function S. Then $\Pr[\sum X_i > s] = S(s/d)$, while the equal split gives $d\,S(s/d)$; whenever the equal split is optimal, the ratio is d. For two independent $\mathrm{Exp}(1)$ variables, $\Pr[X_1 + X_2 > s] = (1+s)e^{-s}$, while $\inf_t (e^{-t} + e^{-(s-t)}) = 2e^{-s/2}$. The ratio $2e^{s/2}/(1+s)$ is unbounded.

c) Monotonicity in the p-box (proof of Theorem 1).: If $X'_e \succeq_{\text{st}} X_e$, there is a coupling with $X'_e \ge X_e$ almost surely. For any copula of (X_g, X_f, X_e), compose it with this monotone coupling to obtain a copula of (X_g, X_f, X'_e) under which the sum is pointwise no smaller. Hence $\overline{p}$ evaluated at the stochastically largest member of the box (CDF $\underline{F}_{\text{env}}$; in the pilot, the 30 kt marginal) dominates $\overline{p}$ at every F_{env} in the box.

Appendix C
Rearrangement Algorithm

For a confidence level α, build an $N \times d$ matrix whose column j holds the quantiles of X_j at the levels $\alpha + (1-\alpha)\,i/N$, for $i = 1, \ldots, N$ in the upper discretisation and $i = 0, \ldots, N-1$ in the lower one. Iterate until the minimum row sum stabilises. In each iteration, rearrange every column so that it is oppositely ordered to the sum of the other columns. The resulting minimum row sum approximates the worst-case value-at-risk $\overline{\mathrm{VaR}}_\alpha(\sum X_j)$ [31], and the conservative of the two discretisations is kept. Then $\overline{p}(s) \le 1 - \sup\{\alpha : \overline{\mathrm{VaR}}_\alpha \le s\}$, found by bisection. The RA's convergence to the sharp bound is not proven in general. For $d = 2$, our implementation (`depbounds.py`) reproduces $\overline{p} = p_U$ to within 0.001 wherever $p_U > 0.002$; below that level, the RA's $1/N$ resolution floor dominates, which is why we report the minimum of the two bounds.

Appendix D
Simulator Constants

Constant	Value	Provenance
γ	1.625 m/s	measured, UAV Attack Dataset [20] (674 samples)
s	4.84 s	measured, artifact (15,392 hops)
L	1.181	artifact, local estimate
step / GNSS rate	20 / 10 Hz	model
PD gains (k_p, k_d)	(1.2, 1.8)	model
drag c_d	$0.35\ \mathrm{s}^{-1}$	model
Dryden h, V	100 ft, 8 m/s	MIL-F-8785C
W_{20} envelope	15–30 kt	declared (C4)
GNSS white / GM	0.5 m / 0.8 m, 60 s	model
accel. noise / bias RW	$0.05\ \mathrm{m/s^2}$ / $0.002\ \mathrm{m/s^2}/\sqrt{\mathrm{s}}$	model (Allan class [110])
anchor σ, T_a	0.3 m, 5 s	model (C2′)
fault probabilities	0.3 / 0.2 / 0.2	model (RflyMAD categories)
outage / jam	0.3 (mean 5 s) / 0.2	model